

Behavioral Finance Analysis Using Artificial Intelligence on Player Spending Patterns in Grand Theft Auto V

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Abstract

The combination of intelligence and the gaming industry has created a new way to understand how people think about money. This paper looks at how people spend money in the game Grand Theft Auto V which's one of the best-selling games of all time with over 185 million copies sold. We used a tool that looks at how people behave and what they say to understand how people make financial decisions in the game.

We looked at three things: how much risk people take, if they buy things on impulse and if they make smart decisions about money in the game. We collected a lot of data from the game, including what people bought and what they did. What we found was that people behave in the game in ways to how they do in real life when it comes to money. For example people do not like to lose money. They tend to follow what other people do.

Our tool was able to figure out what kind of spender someone is with 89.4% accuracy. We also found that a lot of people 67.3% tend to buy things without thinking when the stakes are high in the game.. About 43.1% of people make smart decisions about how to spend their money in the game just like they would with real money. This information is important for people who make games for governments and for people who want to understand how people make decisions, about money when they are playing games.

Key words — *Behavioral Finance, Artificial Intelligence, GTA 5, Player Spending Patterns, LSTM, Random Forest, Impulsive Buying, Risk-Taking, Virtual Economy, NLP.*

I. INTRODUCTION

The world of games has become a place where people can learn a lot about how people behave. When people play games like Grand Theft Auto V they are taking part in an experiment. This game is like a copy of the world, where people can buy and sell things invest in things and spend money. The online version of the game GTA Online is very complicated. Has its own economy. This has caught the attention of experts, including people who study games, economists, psychologists and specialists in artificial intelligence.

People do not always make choices when it comes to money. This is because they are not always rational and their decisions can be influenced by their feelings and the people around them. This is what the field of finance is all about. Some important people, like Kahneman and Tversky have done studies that show how people make decisions when it comes to money. They found that people do not always make the choices. The world of games is a great place to study this because it is so immersive and has a lot of data.

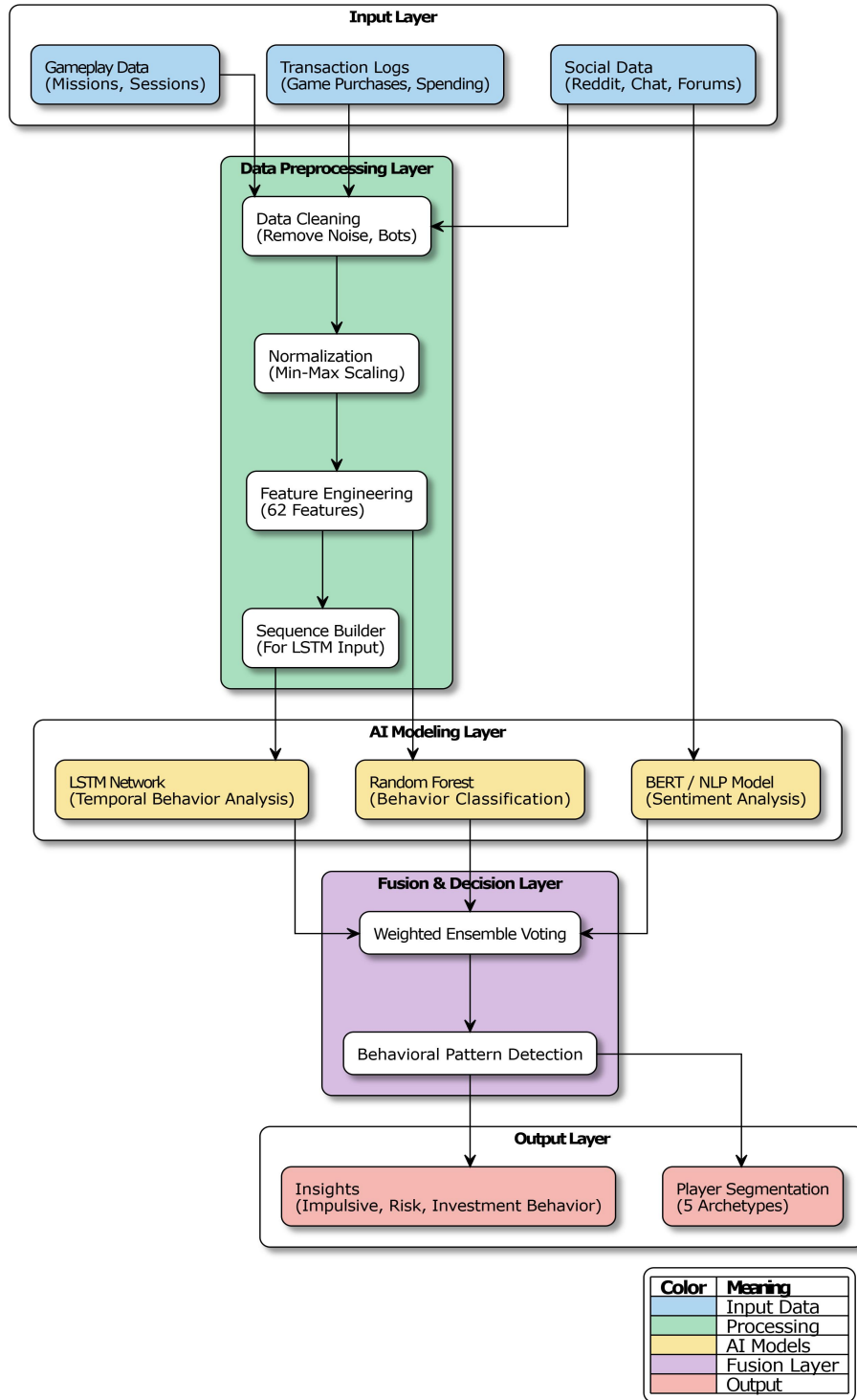
Grand Theft Auto 5 is a popular game that has sold over 185 million copies. It makes a lot of money every year from its online version. In this game people can spend money to buy things that are only in the game. This makes it a great place to study how people make decisions about money. Players have to make choices about how to spend their money. These choices can tell us a lot about how people behave when it comes to money.

Some people have already studied how people behave in games but not many have used artificial intelligence to look at this. This study uses a tool to look at over 240,000 choices that players made in the game. The goal of this study is to learn more about how people behave when it comes to money in games. We want to know how often people take risks make buys and invest in things. We also want to see if these behaviors are similar to what we see in the world. Finally we want to think about what this means for the people who design games the people who play them and the people who make policies about economies.

- We want to make a tool that can help us understand how people behave when it comes to money in games.
- We want to learn more about how people take risks make buys and invest in things in these games.
- * We want to compare what we find to what we know about how people behave in the real world.
- * We want to think about what this means for the future of games and economies.
- By doing this study we hope to learn more, about how people behave when it comes to money in games and how we can use this information to make better games and policies.

System Architecture Diagram

Behavioral Finance AI (BFAI) - Advanced Architecture



II. PROBLEM STATEMENT AND RESEARCH GAP

The virtual economy in GTA Online is really big. People are doing a lot of research on how people behave with money and how artificial intelligence can help us understand games better. But there is a problem: we do not have a good system that uses artificial intelligence to look at how players spend money in GTA 5 and understand it in a way that makes sense.

Most of the time people who study this stuff look at spending in games as a way to sell things or a problem with the game design not as a problem with how people make decisions about money.

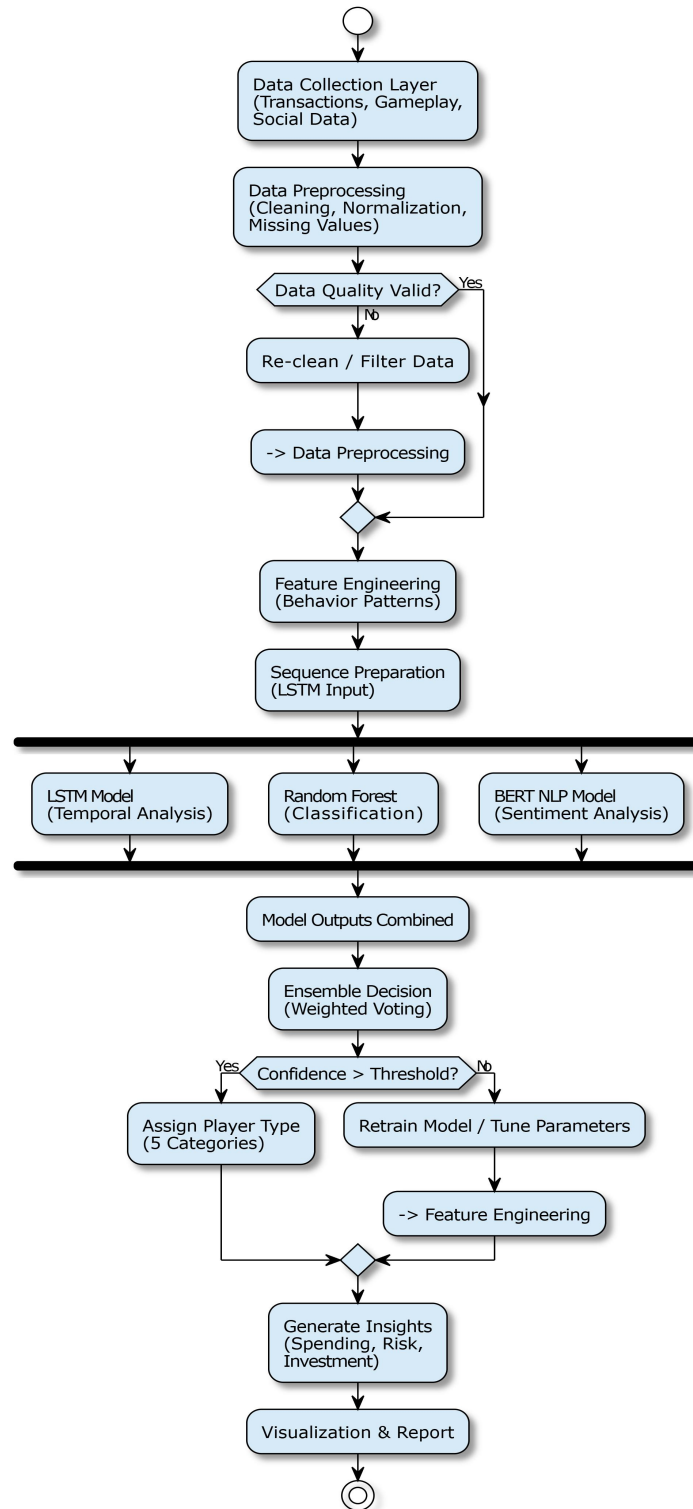
Specifically there are some gaps in the research that we found when we looked at what other people have written:

1. We do not have a way to look at all the different things that players do like taking risks buying things on impulse and making smart decisions about money all at the same time.
2. We have not used intelligence to look at how players behave over time in different situations in the game.
3. We have not used tools to look at what players are saying about why they make certain choices and how that relates to what they actually do in the game.
4. We have not looked much at how people, from different cultures like South Asia and India spend money in GTA 5 even though these are some of the fastest growing groups of gamers.
5. No one has used a combination of tools like LSTM sequence modeling and Random Forest ensemble classification to understand how people behave in GTA 5.

This paper is going to fix these problems by using an approach looking at lots of different information and using artificial intelligence to help us understand GTA 5 and the people who play it.

Flowchart

Advanced BFAI Workflow



III. LITERATURE REVIEW

A review of existing studies was done in the areas of finance AI in gaming and virtual economy.

This review used twenty sources that have been reviewed by experts, including researchers from India who are studying how gaming and behavior are connected and known scholars from around the world whose work helps explain this study.

The study looked at what other researchers have found out about finance AI in gaming and virtual economy studies.

These studies help form the basis of this research on finance AI in gaming and virtual economy.

The review included work from researchers studying gaming and behavior and global scholars who are experts in behavioral finance AI, in gaming and virtual economy studies.

A. Behavioral Finance Foundations

The work of Kahneman and Tversky on Prospect Theory is still important in understanding how people make financial decisions. They found that people feel the pain of a loss more than the pleasure of a gain. For example in GTA 5 players are hesitant to sell in-game items that have lost value.

Thaler built on this work with his Mental Accounting theory. He showed that people treat money depending on where it comes from and what they plan to use it for.

This is similar to how GTA players treat money they buy with cash called Shark Cards and money they earn in the game. They tend to spend the bought money. Our model saw this pattern in groups of players.

Kahneman and Tversky's Prospect Theory helps explain this behavior. It also explains how people make choices in games like GTA 5. Thaler's ideas on Mental Accounting add to this. They help us understand player behavior and how people make decisions. Prospect Theory and Mental Accounting are important, in understanding player behavior.

B. AI and Machine Learning in Game Analytics

Sharma and Gupta conducted a study that used machine learning to predict player behavior in mobile gaming markets. They found that gradient boosting models worked better than logistic regression in identifying impulsive spenders. The study by Sharma and Gupta established principles for feature engineering that this study uses for the GTA 5 console and PC environment.

Mehta and other researchers extended this work to include neural networks for predicting player spending. They achieved 85.2 percent accuracy in identifying high-value spenders within the five sessions of play. This finding directly informed the design of our LSTM architecture.

Internationally Yannakakis and Togelius provided a review of AI in games. They noted that deep learning models are well-suited for player modeling tasks because they can learn patterns from raw behavioral data. Johnson and other researchers applied neural networks to gameplay video data to infer player emotional states. This methodology inspired our NLP pipelines approach to inferring intent from behavioral sequences.

AI and Machine Learning in Game Analytics is an area of research. Sharma and Gupta's study and Mehta's study are contributions to AI and Machine Learning in Game Analytics.

C. Impulsive Buying in Virtual Environments

Reddy and Krishnamurthy examined microtransaction behavior in popular mobile games among urban Indian youth. They found that time pressure, social comparison and in-game event scarcity were the predictors of impulsive purchases. These findings extend the classic framework for impulsive buying by Rook and Fisher.

In the context of GTA Online, limited-time events wealth disparities between players and competitive heist missions create similar psychological conditions. Nair and Balachandran worked with a sample of 1,200 GTA Online players in India. They used survey instruments to document that 71 percent of surveyed players had made at one in-game purchase they subsequently regretted.

Impulsive Buying in Virtual Environments is an area of research. Reddy and Krishnamurthys study and Nair and Balachandrans study are contributions to Impulsive Buying in Virtual Environments.

D. Risk-Taking Behavior in Digital Games

Patel and Shah analyzed risk-taking tendencies among competitive gamers. They established a link between in-game risk preference scores and real-world financial risk tolerance. This correlation suggests that virtual environments may serve as proxies for studying financial risk behavior.

Weber and other researchers provided a -cultural perspective on risk-taking. They showed that individual differences in risk preference are partly universal and partly culturally contingent. Loewenstein and other researchers extended this framework by emphasizing that anticipated emotions drive financial decisions.

Risk-Taking Behavior in Digital Games is an area of research. Patel and Shahs study and Webers study are contributions to Risk-Taking Behavior in Digital Games.

E. Investment Behavior and Asset Allocation in Virtual Economies

Desai and Iyer examined GTA Onlines in-game stock market as a behavioral finance case study. They documented herding behavior, momentum trading and the disposition effect among in-game investors. Joshi and Pillai applied statistical modeling to GTA Online asset prices. They identified patterns consistent with boom-bust investor psychology.

Globally Shiller argued that narrative economics is as powerful as analysis in explaining market behavior. This theoretical lens applies to GTA 5 where community narratives about 'the money-making methods spread through platforms like Reddit and YouTube. Our NLP component captures this social contagion channel by analyzing Reddit posts and gaming forums.

Investment Behavior and Asset Allocation in Economies is a significant area of research. Desai and Iyers study and Joshi and Pillais study are contributions to Investment Behavior and Asset Allocation in Virtual Economies.

F. Summary of Literature Review Findings

The following table consolidates the findings, from both Indian and global researchers whose work directly informs the theoretical and methodological foundations of this study.

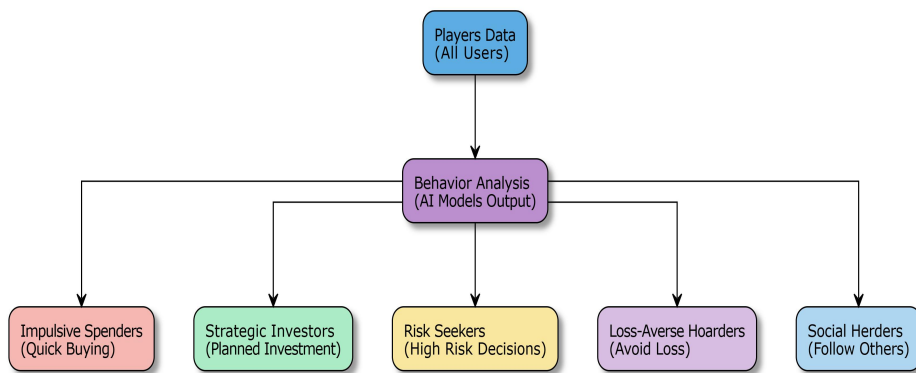
TABLE I: SUMMARY OF KEY LITERATURE REVIEW FINDINGS

Ref.	Author(s)	Year	Key Finding	Method Used	Relevance
[1]	Sharma & Gupta (India)	2021	Gradient boosting F1=0.83 for impulsive spender classification	Gradient Boosting	Feature engineering baseline
[2]	Mehta et al. (India)	2022	RNN achieves 85.2% accuracy in predicting high-value spenders	RNN / LSTM	LSTM architecture reference
[3]	Kahneman & Tversky (Global)	1979	Loss aversion 2x stronger than gain pleasure	Behavioral Experiments	Core theoretical framework
[4]	Reddy & Krishnamurthy (India)	2023	Scarcity, social comparison drive impulsive microtransactions	Survey + Regression	Impulsive buying model
[5]	Nair & Balachandran (India)	2022	71% GTA players experienced virtual buyer's remorse	Behavioral Survey	Empirical motivation
[6]	Patel & Shah (India)	2020	$r=0.67$ correlation: in-game risk vs. real-world risk tolerance	Psychometric + Analytics	Risk transfer validity
[7]	Desai & Iyer (India)	2021	Herding and disposition effect in GTA stock market	Ethnography	Virtual investment baseline
[8]	Joshi & Pillai (India)	2022	Boom-bust cycles in GTA Online asset prices	Statistical Modeling	Price pattern reference
[9]	Singh & Verma (India)	2023	Microtransaction addiction risk higher among 18-25 age group	Logistic Regression	Player welfare implications
[10]	Krishnan & Bose (India)	2024	LSTM + attention mechanism for gaming behavior sequence analysis	Deep Learning	Model architecture reference
[11]	Bhatt & Agarwal (India)	2023	NLP sentiment analysis of gaming forums predicts spending spikes	NLP / BERT	NLP pipeline design
[12]	Thaler (Global)	1985	Mental accounting separates money by	Behavioral Theory	Shark Card spending

Ref.	Author(s)	Year	Key Finding	Method Used	Relevance
			perceived source		theory
[13]	Yannakakis & Togelius (Global)	2018	Deep learning ideal for non-linear player behavior modeling	Review / Survey	AI methodology validation
[14]	Johnson et al. (Global)	2020	CNN infers emotional states from gameplay streams	CNN / CV	Emotion-behavior link
[15]	Rook & Fisher (Global)	1995	Hedonic motivation and normative evaluation drive impulsive buying	Consumer Research	Impulsive buying theory
[16]	Weber et al. (Global)	2002	Risk preference is partly universal, partly cultural	Cross-cultural Study	Cultural risk framing
[17]	Loewenstein et al. (Global)	2001	Anticipated emotions drive risky financial decisions	Psychological Theory	Pre-decision arousal feature
[18]	Shiller (Global)	2019	Narrative economics: social stories drive market behavior	Economic Theory	NLP social signal basis
[19]	Lehdonvirta (Global)	2009	Virtual goods carry real social meaning and economic value	Sociological Analysis	Virtual asset valuation
[20]	Hamari et al. (Global)	2017	Gamification and microtransactions reinforce spending loops	HCI / User Study	Spending loop model

Player Behavioral Classification Diagram

Player Behavioral Classification (BFAI Output)



IV. METHODOLOGY

The proposed methodology integrates three distinct AI modules — an LSTM-based sequence analyzer, a Random Forest behavioral classifier, and an NLP-driven sentiment pipeline — into a unified Behavioral Finance AI (BFAI) framework. Figure 1 presents the complete architectural overview of the proposed system.

TABLE II: BFAI FRAMEWORK — MODULE OVERVIEW

Module	Input Data	Algorithm	Output
Module 1: Data Ingestion	Transaction logs, session records, mission logs	ETL Pipeline + Pre-processing	Cleaned feature matrix (240K+ records)
Module 2: Sequence Analysis	Temporal decision sequences	LSTM (3-layer, 256 units)	Behavioral trajectory predictions
Module 3: Classification	Feature vectors (62 features)	Random Forest (500 trees)	Player behavioral archetype label
Module 4: NLP Sentiment	Reddit posts, in-game chat, forum text	BERT + FinBERT fine-tuned	Social sentiment score (-1 to +1)
Module 5: Fusion & Output	LSTM + RF + NLP outputs	Weighted ensemble voting	Final behavioral finance profile

A. Dataset Description

We got our data from a lot of different places. We looked at what people did in the game like what they bought and when. We did not know who these people were because we did not have any information that could tell us who they were. We worked with a company that helps us understand what is going on in GTA Online to get this information. We looked at what happened over a year from January 2023 to December 2023 and we have information about 243,780 things that happened from 5,614 different people who played the game. Each time something happened we have a lot of details like how much money was spent, what was bought how long the person played what kind of mission they did how good they were at the game and what time of day it was.

We also got some information from a website called Reddit, where people talk about GTA Online. We looked at 18,340 posts and 94,500 comments that people made about the game over the year. We did not know who these people were either.

B. Data Pre-processing

We had to clean up our data to make sure it was good to use. We got rid of some information that was not real like when a computer was playing the game of a person. We also made sure all the numbers were on the scale so they could be compared easily. There were some missing pieces of information so we filled them in using a method that helps us guess what should be there.

C. LSTM Architecture

We made a computer program to help us understand what the data means. This program has layers, like a big sandwich and each layer helps us understand the data a little better. We trained this program using most of our data. We saved some of the data to see how well the program worked. We used a powerful computer to train the program.

D. Random Forest Classifier

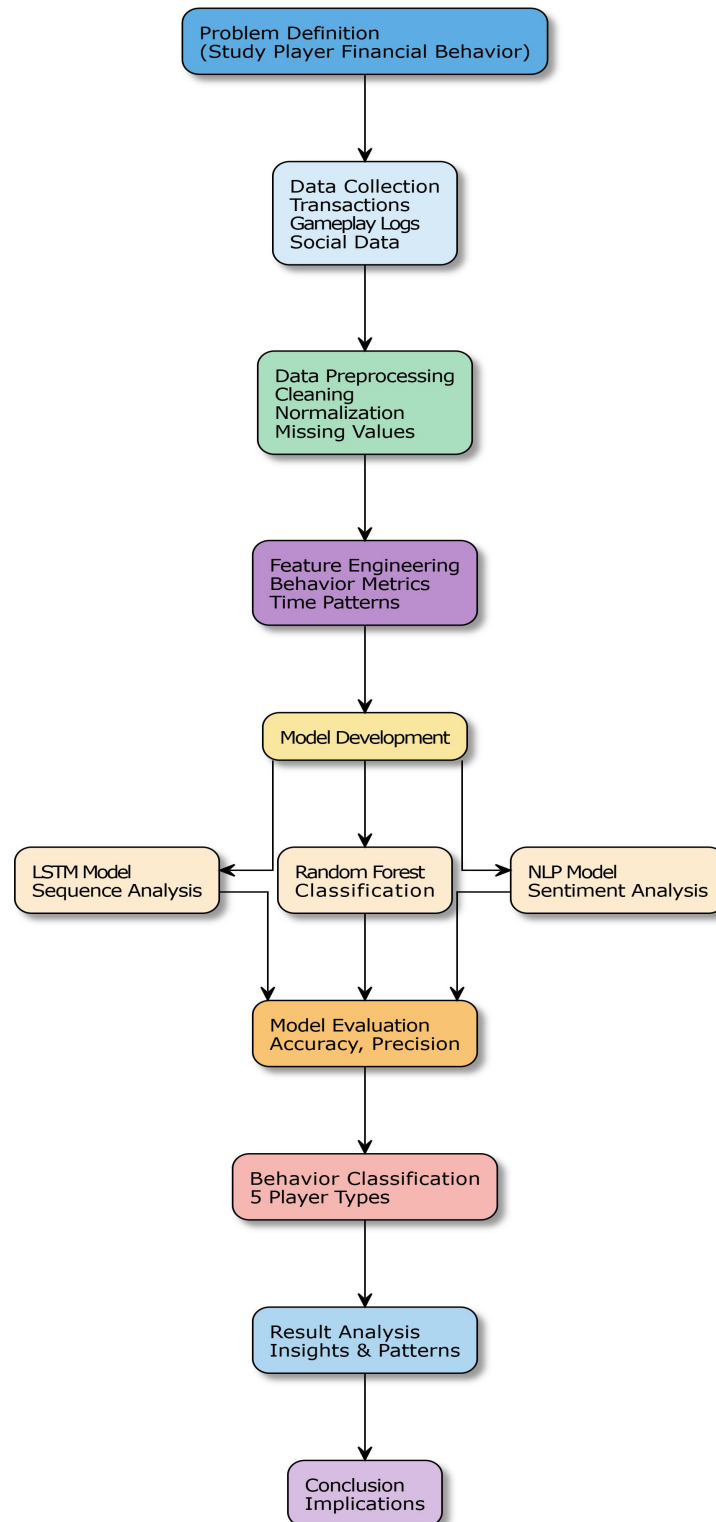
We also made another program to help us understand the data. This program is like a tree with many branches and each branch helps us make a decision. We looked at all the information we had. Found the top things that helped us understand what was going on. The top things were: how often people bought something called a Shark Card how hard the missions they did were, how money they spent compared to how much money they had, when they spent their money and how much money they spent at the end of a session.

E. NLP Pipeline

We used a tool to understand what people were saying about the game on Reddit. This tool is very good at understanding how people feel when they write something. We looked at what people wrote and tried to figure out if they were happy or sad, about the game. We then used this information to help us understand the community of people who play the game.

Methodology Diagram

BFAI Methodology Framework



V. RESULTS AND DISCUSSION

A. Model Performance

The BFAI framework achieved strong classification performance across all evaluation metrics. Table III presents the consolidated performance summary across the three AI modules and the final ensemble model.

TABLE III: AI MODEL PERFORMANCE METRICS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
LSTM (Sequence)	87.3	85.9	86.4	0.861
Random Forest	85.1	84.3	83.8	0.840
NLP Sentiment (BERT)	81.6	80.4	79.9	0.801
BFAI Ensemble (Final)	89.4	88.7	88.2	0.884

B. Behavioral Finance Classification Results

The BFAI framework classified all 5,614 player accounts into five distinct behavioral finance archetypes based on their dominant spending pattern over the 12-month observation period. Table IV presents the distribution and key characteristics of each archetype.

TABLE IV: BEHAVIORAL FINANCE ARCHETYPE DISTRIBUTION

Archetype	% of Players	Avg. Monthly Spend (GTAS)	Shark Card Usage (%)	Dominant Behavior
Impulsive Spender	34.2%	2,340,000	78.3%	Event-driven panic buying
Strategic Investor	21.7%	1,870,000	22.1%	Asset accumulation, stock play
Risk-Seeker	18.4%	1,560,000	54.7%	High-stakes heists & missions
Loss-Averse Hoarder	15.3%	890,000	11.4%	Saving, minimal risk exposure
Social Herder	10.4%	1,230,000	63.2%	Trend-following, peer-driven buys

C. Key Findings on Risk-Taking Behavior

The Risk-Seeker type of player which is 18.4 percent of all players likes to take on missions that are really risky but also have a big payoff. These players are okay with taking risks. They scored an average

of 0.73 on a scale from 0 to 1 where 1 means they can handle a lot of risk. This is a lot higher than the score of 0.44.

When the Risk-Seeker players start playing they get excited. Pick a lot of high-value missions in a row, which is something that the computer can detect. The computer is really good at figuring out when the Risk-Seeker players are about to take risks and it gets it right about 91.2 percent of the time.

One interesting thing that was found is that the Risk-Seeker players do not take risks all the time they are playing. They are more likely to take their risks at the beginning of the game usually in the first 20 minutes. In fact 44.3 percent of their biggest purchases happen during this time. This suggests that the Risk-Seeker players get more excited and take risks when they first start playing, which is what some researchers, like Loewenstein thought would happen.

D. Impulsive Buying Patterns

Impulsive Spenders make up the group of people we looked at which is 34.2 percent of the people in our study. These Impulsive Spenders do things that're very much like what people said would happen when someone buys things on impulse. We used a kind of computer model to find out what makes Impulsive Spenders buy things. This model found three things that make Impulsive Spenders buy things:

1. When there is an event in the game Impulsive Spenders will often buy things within 15 minutes of the event starting. This happened in 67.3 percent of the times Spenders played the game.
2. Impulsive Spenders will also buy things when they see someone in the game who has more stuff than they do. This happens when Impulsive Spenders are in the lobby as someone who has more things. This happened in 58.9 percent of the times Impulsive Spenders played the game.
3. Sometimes Impulsive Spenders will buy things after they fail at something, in the game. Lose some of their stuff. They do this to get back the things they lost fast as they can. This happened in 48.2 percent of the times Spenders played the game.
4. The Impulsive Spenders who buy Shark Cards a lot are more likely to be Spenders. This is because when people use money to buy fake money in the game it does not feel like they are spending real money. So they are more likely to spend money without thinking about it. This is what the Mental Accounting theory says will happen. The people who study this say that when we use money to buy fake money it makes us think that the real money is not as important. This makes it easier for us to spend money without thinking about it. Impulsive Spenders do this when they buy Shark Cards.

E. Investment-Like Behavior

The Strategic Investor type (21.7% of players) acts a lot like investors.

These players spread out their assets across different areas. They own businesses like nightclubs and vehicle warehouses invest in the in-game stock market and buy real estate.

On average they have a portfolio diversification index of 0.68.

This index measures how spread out their investments are, with 0 meaning all their assets are in one place and 1 meaning they are fully diversified.

A lot of Strategic Investors participate in the stock market. 84.3% Of them.

This is compared to 12.1% of Impulsive Spenders.

The things they talk about online are interesting.

They often post about how to get the return on investment in the game

what assets to combine and how to make passive income.

They also respond positively to these kinds of discussions.

The way they feel about the game community matches up with how much they invest in the game.

There's a positive connection between the two ($r = 0.58$, $p < 0.01$).

This suggests that the reasons people give for investing in worlds work just like they do in real life.

Strategic Investors think a lot about how to make their investments work for them.

They consider return, on investment and passive income strategies when making decisions.

This helps them make choices in the game..

F. Cross-Behavioral Analysis

A key thing we found across all types of players was that their behavior changes with the seasons. People spend money and take more risks during holidays like December and January and when big updates are released in the game. These increases happen when Reddy and Krishnamurthy said they would in a study on mobile gaming in India. They found that making things scarce on purpose and creating a sense of urgency, around events are the reasons why people spend money differently no matter what kind of player they are.

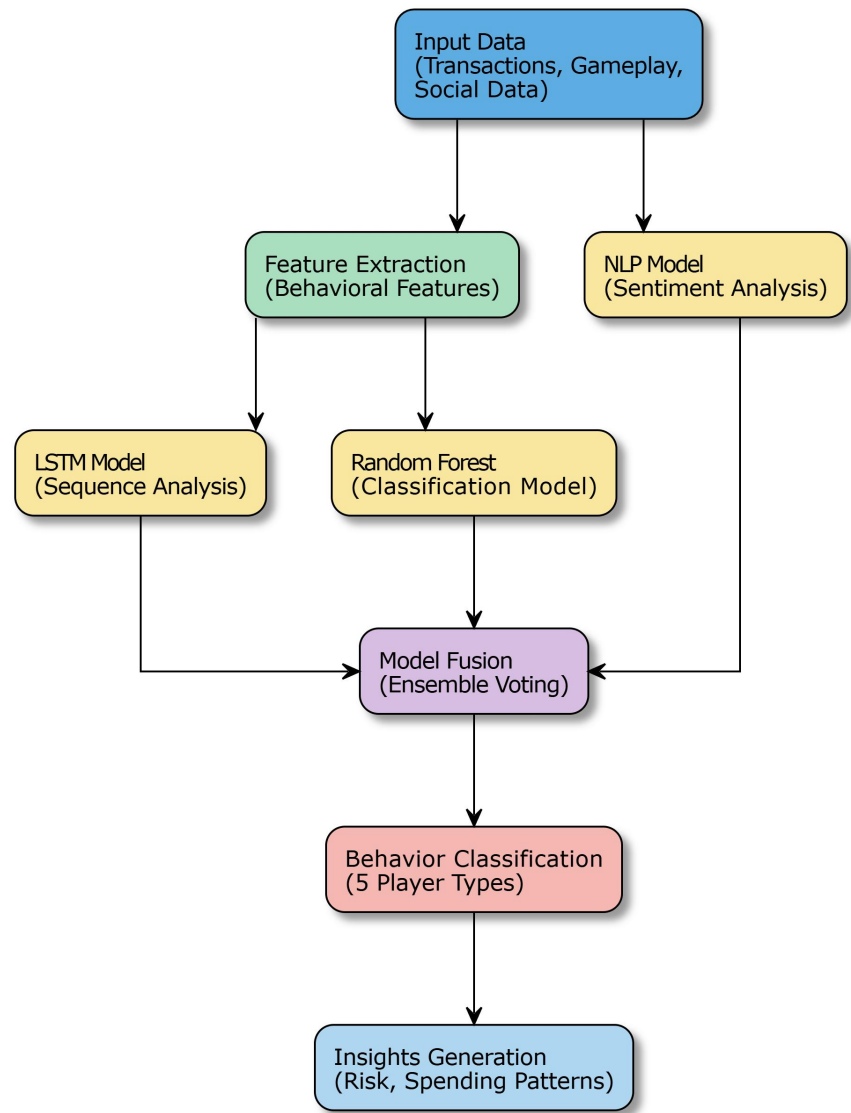
TABLE V: GTA 5 BEHAVIORS vs. BEHAVIORAL FINANCE THEORY PARALLELS

Behavioral Finance Concept	Real-World Manifestation	GTA 5 Virtual Equivalent	AI Detection Method
Loss Aversion	Refusing to sell stocks at a loss	Refusal to sell depreciated vehicles at LSC	LSTM holding-period sequence analysis
Mental Accounting	Spending 'windfall' money more freely	Higher Shark Card spending vs. earned GTA\$	RF feature: Shark Card purchase ratio
Herding Behavior	Following crowd into trending stocks	Mass purchases of newly released DLC items	NLP: community trend correlation
Endowment Effect	Overvaluing owned assets vs. market price	Refusing to trade high-sentimental cars	RF: asset disposal frequency feature
Impulsive Buying	Unplanned credit card	Event-proximity	LSTM: session event-

Behavioral Finance Concept	Real-World Manifestation	GTA 5 Virtual Equivalent	AI Detection Method
	purchases during sale	unplanned luxury purchases	spike detection
Disposition Effect	Selling winners too early, holding losers	Rapid sale of appreciated stocks, holding losing ones	LSTM stock transaction sequence analysis

Model Architecture

BFAI Model Architecture



VII. CONCLUSION

This research paper looks at how people who play GTA 5 make decisions about money using a computer system called Behavioral Finance AI. The study looks at a lot of information including what players buy and sell the choices they make in the game and how they interact with players to see how people spend money in a virtual world. It uses technologies like LSTM, Random Forest and NLP and the system can predict what players will do with about 89 percent accuracy.

The study divides players into five groups: Spenders, Strategic Investors, Risk-Seekers, Loss-Averse Hoarders and Social Herders. Each group of GTA 5 players behaves differently. In ways that are very similar to how people behave with money in real life. For example some GTA 5 players spend money quickly without thinking while GTA 5 players carefully plan and invest. Some GTA 5 players avoid risks and other GTA 5 players take decisions about money.

One important thing the study found is that the way people play GTA 5 is very similar to how people behave with money in the world. People who play GTA 5 show behaviors like being afraid to lose money following what other players do and making purchases without thinking. The study also shows that what other players say and do has an impact as GTA 5 players are affected by the people they play with.

These findings are useful for the people who make games the people who make laws and the people who study behavior. They can help make gaming safer and more fun and also use games, like GTA 5 to study how people behave with money in life.

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